1. Identity

Document	STAR-PKG-STAR-EULER-EN · v2.2 · 2026-11-02
Package	star-euler
Version	0.1.0
Capability	star.attitude.euler
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_euler

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_euler-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_euler-0.1.0-py3-none-any.whl	017cd461611a8e50d9e4c825ed8c887dcdc4a523294c41a9397d0e1d05eb8194
star_euler.py	a43e4c7faa5b0fbc71e02a9076d389eda37416bf772118e52d512fb30f371201
star_euler-0.1.0-source.zip	026884d5e7a02f0397973dc9fb2f8994c27c03c7998b7c1ae68e079c6e2dc2eb

5. What it does, and its limits

Rotation matrix from three Euler angles and back for the six Tait-Bryan and six proper sequences, with one stated convention (active, intrinsic), gimbal lock detected and reported; agrees with SciPy and NAIF SPICE to 6e-16 on all twelve sequences.

No extrinsic (fixed-axis) sequences as a separate option: reverse the letters and the angles to get them. No angular rates or kinematic equations. Within about 1e-7 degrees of gimbal lock the first and third angles are not separable and only their combination is returned.

- Convention: ACTIVE, INTRINSIC rotations: `to_dcm("ZYX", a1, a2, a3) = Rz(a1) Ry(a2) Rx(a3)` (turn about Z, then about the new Y, then about the new X). This is SciPy's upper-case sequence; NAIF SPICE `eul2m` returns the transpose.
- `from_dcm(seq, R)`: middle angle in [-90, 90] (Tait-Bryan) or [0, 180] (proper), the others in (-180, 180]. At gimbal lock the third angle is returned as 0 and the first carries the whole rotation; `is_singular` reports it.
- Measured against SciPy and SPICE for all twelve sequences (720 regular cases, 24 at gimbal lock): matrices within 6e-16 (bit-identical to SPICE), angles within 4e-13 degrees, matrices rebuilt at gimbal lock within 6e-16.
- Every function returns finite floats or raises `ValueError`: a sequence that is not one of the twelve (lower case included: it is not guessed), non-numeric, boolean, NaN or infinite input, a matrix of the wrong shape, not orthonormal to 1e-9, or a reflection.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before

using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_euler_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-euler}: Euler angles for the twelve sequences, one convention},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_euler/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts